BCGFMamba: Bottleneck Context and Gated Fusion for Postoperative Glioma Segmentation

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Abstract. Postoperative glioma segmentation remains challenging because residual tumor regions often exhibit blurred boundaries, fragmented morphology, and substantial appearance variability after surgical intervention. In this work, we present BCGFMamba, a lightweight U-Mamba-based framework for postoperative glioma segmentation. The proposed method introduces two localized architectural enhancements on top of a pretrained U-Mamba backbone: a Bottleneck Context Refinement (BCR) module for enriching the deepest representation with multi-scale contextual cues, and a Gated Skip Refinement (GSR) module for suppressing noisy high-resolution skip features before decoder fusion. To adapt the modified architecture efficiently, we further adopt a staged training strategy consisting of baseline training, module adaptation, and joint refinement of the bottleneck Mamba block and its adjacent decoder stages. Experiments on BraTS 2024 show that BCGFMamba improves Dice from 0.8148 to 0.8275. Ablation results further show that module-adaptation-only variants remain at the baseline level, whereas the joint refinement stage consistently lifts BCR-only, GSR-only, and full BCGFMamba variants to a similar performance range around 0.8275–0.8278. These results indicate that lightweight bottleneck context enhancement and gated skip fusion can be effectively transferred to postoperative glioma segmentation when paired with controlled refinement.

Keywords: Postoperative glioma segmentation · Mamba · Gated fusion · Context refinement

1 Introduction

Accurate postoperative glioma segmentation is crucial for treatment assessment and subsequent clinical decision-making, yet it remains substantially more challenging than preoperative tumor segmentation. Residual tumor regions after surgery often exhibit irregular morphology, fragmented appearance, and weak contrast against surrounding postoperative changes. These properties make reliable delineation difficult even for strong volumetric segmentation backbones. Recent postoperative glioblastoma studies have also emphasized that many segmentation pipelines developed for preoperative MRI do not transfer reliably to the postoperative setting without task-specific adaptation [1,2]. Recent transformer- and state-space-based models have improved long-range dependency modeling,

but practical segmentation quality still depends on how effectively they aggregate deep contextual information and selectively fuse high-resolution encoder features during decoding. U-Mamba is a strong representative of this line of work [4], but its bottleneck representations can still benefit from stronger context refinement, and its decoder skip fusion can still introduce noisy postoperative background responses.

To address these limitations without redesigning the entire backbone, we propose BCGFMamba. The method introduces two localized components on top of a U-Mamba encoder-decoder architecture. First, a Bottleneck Context Refinement (BCR) module is inserted after the Mamba bottleneck to enrich the deepest feature representation using multi-scale contextual aggregation inspired by DeepLab-style design principles [3]. Second, a Gated Skip Refinement (GSR) module is applied to the highest-resolution decoder skip pathways following the intuition of gated feature selection in Attention U-Net [5]. These two modifications target complementary weaknesses of postoperative glioma segmentation: insufficient semantic context at the bottleneck and noisy high-resolution fusion near the output side.

We further introduce a staged training strategy to adapt the modified architecture from a pretrained baseline checkpoint. The full process contains three phases: baseline training on the target dataset, module adaptation of the newly introduced components, and joint refinement of the bottleneck Mamba block together with the adjacent decoder stages. Rather than unfreezing the full backbone, we only relax the trainable set around the bottleneck during the final phase. This design reflects the practical observation that lightweight architectural modifications can require controlled transfer to become effective on postoperative glioma data.

1.1 Related Work

Our design is most closely related to four lines of work. First, U-Mamba-style architectures demonstrate the value of state-space modeling for volumetric biomedical segmentation [4]. Second, recent postoperative glioma and glioblastoma studies emphasize that residual tumor delineation requires tailored modeling and task-specific training strategies rather than direct reuse of generic brain tumor pipelines [1,2]. Third, DeepLab-style atrous context modeling showed that parallel dilated convolutions are effective for enriching multi-scale semantic representations [3]. Fourth, gating and lightweight residual enhancement modules have been widely used to improve feature selection and multi-scale fusion in modern encoder-decoder segmentation networks [5,6]. BCGFMamba combines these intuitions in a U-Mamba-based framework by placing context refinement at the bottleneck and gated filtering at the highest-resolution skip pathways, then adapting the modified network through staged refinement rather than full retraining.

The contributions of this paper are threefold. First, we propose BCGFMamba, a lightweight U-Mamba-based framework that couples Bottleneck Context Refinement with Gated Skip Refinement for postoperative glioma segmen-

tation. Second, we introduce a staged training protocol consisting of baseline training, module adaptation, and joint refinement of the bottleneck and adjacent decoder stages. Third, on BraTS 2024, we show that the full BCGFMamba design improves Dice from 0.8148 to 0.8275, while ablation experiments indicate that the decisive empirical factor is the final joint refinement stage: module-adaptation-only variants stay near the baseline, whereas joint refinement consistently lifts BCR-only, GSR-only, and full BCGFMamba variants to a similar improved range. These results support the claim that lightweight modular upgrades can be effectively transferred to postoperative glioma segmentation when optimization is handled carefully.

2 Method

2.1 Overview

BCGFMamba is a U-Mamba-based segmentation framework for postoperative glioma segmentation. The complete pipeline consists of three phases: baseline training, module adaptation, and joint refinement. During baseline training, a U-Mamba backbone is first trained on the target postoperative dataset and provides the initialization checkpoint for the proposed method. Let $\mathbf{x}$ denote the input postoperative T1ce MRI volume and let $\{\mathbf{s}_l\}_{l=1}^{L}$ be the multi-scale encoder features, where $\mathbf{s}_L$ is the deepest feature map. The pretrained backbone extracts hierarchical encoder features, applies a Mamba bottleneck transformation to $\mathbf{s}_L$, and reconstructs the prediction through a U-shaped decoder with deep supervision heads inherited from the baseline design.

BCGFMamba preserves this overall topology but introduces two localized modifications. First, a Bottleneck Context Refinement (BCR) module is inserted immediately after the bottleneck Mamba layer. Second, Gated Skip Refinement (GSR) modules are placed on the two highest-resolution decoder skip pathways. These modifications target two common failure modes of postoperative glioma segmentation: insufficient multi-scale semantic context at the deepest stage and noisy high-resolution skip fusion around the output side. The complete framework and the staged optimization flow are summarized in Fig. 1.

2.2 Bottleneck Context Refinement

The BCR module is designed to strengthen multi-scale context modeling without changing the surrounding backbone. Let $\mathbf{b} \in R^{C\times D_b\times H_b\times W_b}$ denote the bottleneck feature after the Mamba layer. BCR applies several parallel context branches: a $1\times1\times1$ convolution branch for local channel mixing, three $3\times3\times3$ dilated convolution branches with dilation rates 1, 2, and 3, and a global pooling branch for image-level context. Their outputs are concatenated and projected back to the original bottleneck dimension:

$$\mathbf{b}' = \mathrm{Proj}\left([\phi_{1\times1}(\mathbf{b}) \,\|\, \phi_{d=1}(\mathbf{b}) \,\|\, \phi_{d=2}(\mathbf{b}) \,\|\, \phi_{d=3}(\mathbf{b}) \,\|\, \phi_{\mathrm{gp}}(\mathbf{b})]\right), \tag{1}$$

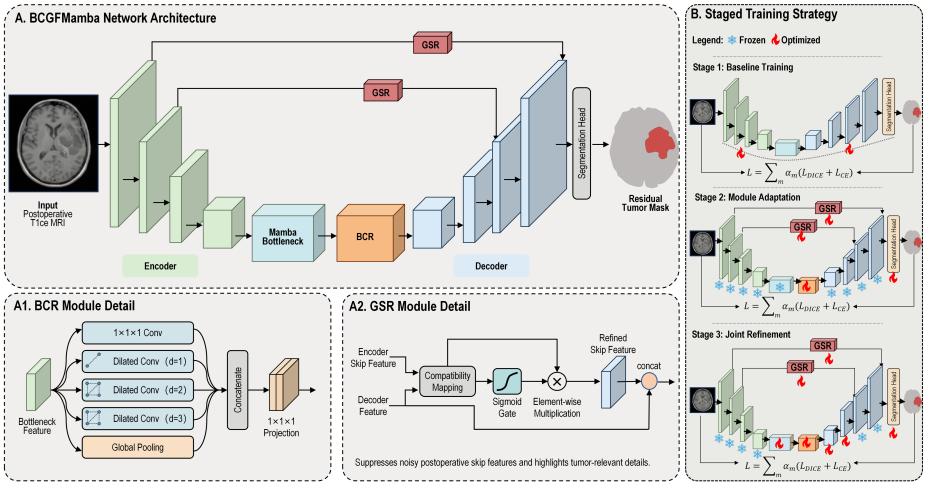


Fig. 1. Overview of BCGFMamba for postoperative glioma segmentation. Panel A shows the U-Mamba-based network architecture with BCR after the bottleneck Mamba block and GSR on the two highest-resolution skip pathways. Panel A1 details the BCR module with parallel $1 \times 1 \times 1$ convolution, dilated-convolution, and global-pooling branches followed by concatenation and projection. Panel A2 details the GSR module, where encoder skip features are gated by decoder-conditioned compatibility mapping and then concatenated with the decoder feature before the decoder block. Panel B illustrates the three-stage optimization protocol consisting of baseline training, module adaptation, and joint refinement, together with the frozen and optimized components and the shared deep-supervision loss used across stages.

where $\phi_{\text{gp}}(\mathbf{b})$ denotes global average pooling followed by $1 \times 1 \times 1$ projection and trilinear upsampling, $\|$ denotes channel concatenation, and $\text{Proj}(\cdot)$ is a final $1 \times 1 \times 1$ projection. This block forms the **BCR** component.

The motivation is straightforward. Although the Mamba bottleneck already captures long-range dependencies, it does not explicitly enforce multi-scale spatial aggregation in the same way as dedicated context modules. By placing BCR at the bottleneck, we enrich the deepest representation with complementary receptive fields before decoder reconstruction, while avoiding invasive modifications to the full network.

2.3 Gated Skip Refinement

High-resolution skip fusion is essential for recovering boundary detail, but it can also introduce irrelevant activations from early encoder stages, especially in postoperative MRI where cavity and treatment-related changes can resemble residual lesion patterns. To mitigate this issue, we place GSR modules on the two highest-resolution skip paths. For a decoder feature $\mathbf{d}_l$ and its corresponding

encoder skip feature $\mathbf{s}_l$, GSR computes an attention mask

$$\mathbf{a}_l = \sigma\left(\psi\left(\delta\left(W_d\mathbf{d}_l + W_s\mathbf{s}_l\right)\right)\right), \tag{2}$$

where W_d and W_s are $1 \times 1 \times 1$ projections, $\delta(\cdot)$ denotes ReLU, $\psi(\cdot)$ maps the intermediate feature to a single-channel response, and $\sigma(\cdot)$ is the sigmoid function. The filtered skip feature is then obtained as

$$\tilde{\mathbf{s}}_l = \mathbf{s}_l \odot \mathbf{a}_l, \tag{3}$$

and concatenated with $\mathbf{d}_l$ before the decoder block. This block forms the **GSR** component.

We restrict GSR to the highest-resolution decoder stages rather than applying it to all scales. The rationale is that the highest-resolution skip features are the most important for boundary recovery, but they are also the most likely to carry redundant texture and postoperative background variation. Selective gating at these stages therefore offers a favorable trade-off between added complexity and potential benefit.

2.4 Staged Training Strategy

Directly retraining the modified network from scratch is not the most efficient way to evaluate localized architectural modifications on top of a strong pretrained baseline. We therefore adopt a staged training strategy that starts from a pretrained baseline checkpoint and progressively relaxes the trainable parameter set.

Stage 1: Baseline training. We first train the baseline U-Mamba backbone on the target postoperative glioma dataset and select its best checkpoint as the initialization source for BCGFMamba. This stage establishes a stable feature prior on the target domain before introducing any additional modules.

Stage 2: Module adaptation. We initialize the proposed model from the pretrained baseline checkpoint using non-strict weight loading. During this stage, we optimize only the newly introduced modules together with the segmentation heads. For BCGFMamba, this means the BCR block, the GSR modules, and the prediction heads are trainable, while the inherited backbone parameters remain frozen. This stage allows the newly added components to adapt to the pretrained feature distribution without destabilizing the baseline representation.

Stage 3: Joint refinement. After the new modules stabilize, we unfreeze the bottleneck Mamba layer together with the decoder stages adjacent to the bottleneck for joint refinement. Importantly, the encoder remains frozen. This stage enables co-adaptation between the pretrained bottleneck representation, the coarse-to-mid-resolution decoder pathway, and the newly added modules, while still avoiding full-network retraining. In practice, the staged protocol serves two purposes: it provides a stable optimization path for lightweight architecture surgery on a

pretrained model, and it creates an interpretable ablation setting in which we can separate the effect of module insertion from the effect of subsequent joint adaptation.

2.5 Training Objective

All three stages optimize the same segmentation objective, but the trainable parameter set differs across stages. In our implementation, the basic segmentation loss is the sum of soft Dice loss and cross-entropy loss:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{CE}}. \tag{4}$$

Because the decoder uses deep supervision, predictions are generated at multiple decoder scales. The final training loss is therefore

$$\mathcal{L} = \sum_{m=1}^{M} \alpha_m \mathcal{L}_{\text{seg}} \left(\hat{\mathbf{y}}^{(m)}, \mathbf{y}^{(m)} \right), \tag{5}$$

where $\hat{\mathbf{y}}^{(m)}$ and $\mathbf{y}^{(m)}$ denote the prediction and supervision target at scale m, and α_m is the corresponding deep-supervision weight. In our training setup, higher-resolution outputs receive larger weights, whereas the coarsest auxiliary output is ignored. This shared objective ensures that performance differences across BCR-only, GSR-only, and full BCGFMamba variants are attributable to architecture and trainable-stage design rather than to a changed loss formulation.

3 Experiments and Results

3.1 Datasets

BraTS 2024 is used in this study. In the paper, we focus on postoperative residual tumor segmentation from T1ce MRI and report results on this dataset. All quantitative results in the present manuscript are obtained under a shared preprocessing, training, and validation protocol so that the comparison isolates architectural and optimization differences rather than data-processing differences.

3.2 Implementation Details

All experiments are implemented in PyTorch on Ubuntu Linux servers equipped with Intel Xeon Gold 6230 CPUs and Tesla V100-SXM2-32GB GPUs. U-Mamba is used as the baseline backbone throughout all comparisons. Before training, the postoperative MRI volumes are resampled to a unified training resolution and intensity-normalized. Unless otherwise stated, all models are trained with a batch size of 1, an initial learning rate of 10^{-2}, weight decay of 3×10^{-5}, and 200 epochs. Deep supervision is enabled for all variants.

Table 1. Comparison with representative baseline methods on BraTS 2024.

Method	Dice	IoU
U-Net	0.519	0.401
U-Net 3+	0.469	0.353
DeepResU-Net	0.465	0.352
nnU-Net	0.728	0.589
nnFormer	0.728	0.585
SAM-Med3D	0.731	0.571
U-Mamba	0.815	0.698
BCGFMamba (ours)	**0.828**	**0.715**

The complete training pipeline consists of three stages. In Stage 1, the U-Mamba backbone is trained on BraTS 2024 and provides the initialization checkpoint for subsequent transfer. In Stage 2, only the newly introduced modules and the segmentation heads are optimized. In Stage 3, the bottleneck Mamba layer and the decoder stages adjacent to the bottleneck are further unfrozen for joint refinement, while the encoder remains frozen. This protocol is shared across the BCR-only, GSR-only, and full BCGFMamba variants so that the ablation isolates architectural differences rather than training-schedule differences. The training objective is identical across all stages and follows Section 2.5. Under this setup, the comparison is controlled at four levels: backbone, initialization, optimization recipe, and loss formulation.

3.3 Main Results

Table 1 summarizes the main comparison on BraTS 2024 using representative baseline methods reported in the reference postoperative glioma segmentation study together with our current results. BCGFMamba achieves a Dice of 0.8275 and an IoU of 0.715, improving over the U-Mamba baseline scores of 0.8148 Dice and 0.698 IoU. It also exceeds the representative CNN-, transformer-, and foundation-model-based baselines listed in the table. This result supports the practical value of the proposed design: localized bottleneck and skip refinements, when paired with staged optimization, can improve a strong postoperative segmentation backbone without redesigning the full architecture.

3.4 Ablation Study

The ablation study is organized around two questions: whether bottleneck context refinement and gated skip refinement each provide a measurable standalone gain, and whether the final joint refinement stage is necessary to unlock the benefit of the modified architectures. Table 2 shows a clear pattern. After Stage 2 module adaptation, all variants remain near the baseline (BCR-only: 0.814791, GSR-only: 0.814816, BCGFMamba: 0.814815), indicating that module insertion alone does not immediately improve performance. After Stage 3 joint refinement,

Table 2. Ablation study of the proposed modules and staged fine-tuning on BraTS 2024.

Variant	Module Adaptation	Module Adaptation + Joint Refinement
BCR-only	0.814791	**0.827831**
GSR-only	0.814816	0.827735
BCGFMamba	0.814815	0.827515

however, all three modified variants improve substantially: BCR-only reaches 0.827831, GSR-only reaches 0.827735, and full BCGFMamba reaches 0.827515.

Two conclusions follow from these results. First, both BCR and GSR are viable plug-in modifications once the pretrained backbone is allowed to co-adapt with them during the joint refinement stage. Second, the optimization schedule is more influential than the mere number of inserted modules: without Stage 3, none of the variants departs from the pretrained baseline in a meaningful way. The final BCGFMamba configuration stays in the top-performing tier while providing the most complete bottleneck-and-skip refinement design for the paper's final model definition.

From a reporting perspective, these ablations also clarify the intended paper claim. The main contribution is not that every added component independently guarantees monotonic improvement in a frozen setting. Instead, the contribution is that lightweight modifications around the bottleneck and high-resolution skip fusion can be transferred effectively to a pretrained U-Mamba backbone when the refinement scope is chosen carefully.

3.5 Qualitative Analysis

Figure 2 presents three representative BraTS 2024 cases in a reference-style side-by-side comparison. For each case, we compare the input image, the ground truth, the pretrained U-Mamba baseline, and the final BCGFMamba prediction. All masks are rendered in red, and the green boxes highlight regions where the proposed method improves residual tumor extent recovery or local boundary delineation relative to the baseline. The first case emphasizes a strong correction on a difficult baseline failure, while the second and third cases show that the improvement pattern also persists in moderate- and high-quality predictions.

4 Discussion

The empirical pattern in this study is more informative than any single summary result. All module-adaptation-only variants remain essentially indistinguishable from the pretrained baseline, while all jointly refined variants improve to a clearly higher performance range. This indicates that for postoperative glioma segmentation, the benefit of localized architectural modification depends strongly on whether the pretrained bottleneck representation and decoder pathway are allowed to co-adapt with the inserted modules. In other words, module quality

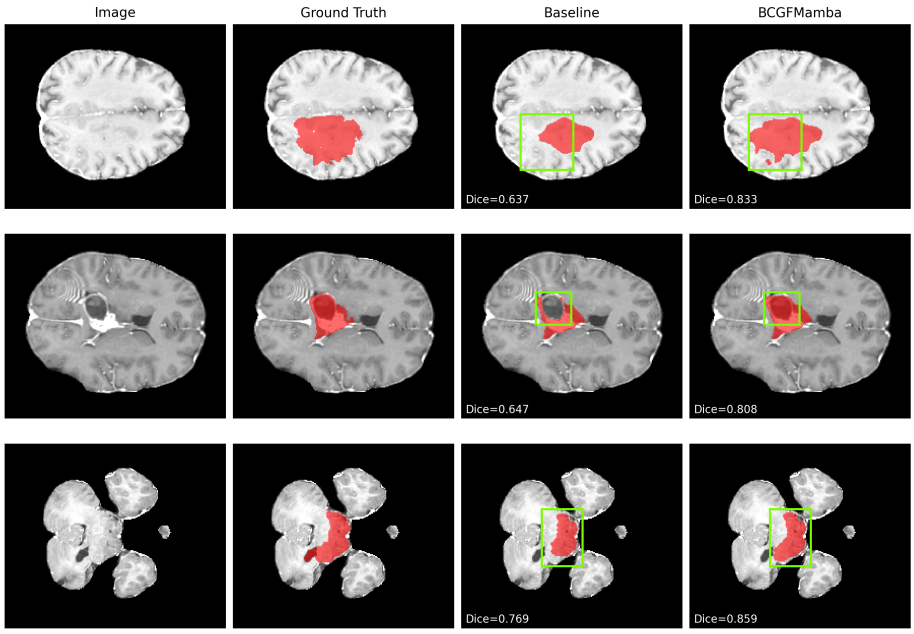


Fig. 2. Qualitative comparison on representative BraTS 2024 postoperative glioma cases. Each row shows one case, and each column shows the input image, ground truth, pretrained U-Mamba baseline, and BCGFMamba prediction. Red overlays denote the target region, and the green boxes mark areas where BCGFMamba improves residual tumor recovery or local shape consistency relative to the baseline.

and optimization scope are coupled: a reasonable plug-in block may still appear ineffective if evaluated only under a largely frozen backbone.

This observation is relevant in the context of recent postoperative glioma and glioblastoma segmentation studies. Prior work has highlighted that postoperative segmentation differs substantially from the preoperative setting because of surgically induced cavities, treatment-related enhancement, and ambiguous residual boundaries [1,2]. Many recent postoperative pipelines therefore rely on multiparametric MRI and broader label sets such as enhancing tumor, edema, and surgical cavity [2]. By contrast, our setting focuses on residual tumor segmentation from postoperative T1ce MRI with U-Mamba as the backbone. Under this more targeted setup, the present results suggest that carefully designed bottleneck context aggregation and gated skip filtering can still provide measurable gains without replacing the underlying backbone.

The ablation results also warrant a nuanced interpretation of the final model choice. The BCR-only variant achieves the numerically highest Dice, but the margin over full BCGFMamba is small. We therefore retain BCGFMamba as the primary model reported in the paper because it unifies bottleneck refinement and skip refinement in a coherent architecture while remaining within the

same performance band. This framing is methodologically defensible as long as the paper reports the ablation transparently, which we do. The qualitative comparisons are also consistent with this positioning: the selected BraTS 2024 cases show that the full model recovers missing residual regions and improves local shape coherence relative to the pretrained baseline, especially when the baseline under-segments difficult postoperative targets.

Several limitations should be acknowledged. First, the current evidence is based on performance on BraTS 2024 only, rather than a broader multi-center benchmark. Second, the study focuses on Dice and does not yet report boundary-sensitive metrics or inference-cost comparisons. These limitations do not invalidate the current findings, but they define the next steps needed to strengthen the final submission.

5 Conclusion

We presented BCGFMamba, a lightweight U-Mamba-based framework for postoperative glioma segmentation that combines Bottleneck Context Refinement, Gated Skip Refinement, and staged training. By modifying only the bottleneck and the highest-resolution skip fusion stages, the proposed method preserves the overall backbone structure while yielding a clear Dice improvement on BraTS 2024, from 0.8148 to 0.8275. The completed ablation study shows that module-adaptation-only variants remain close to baseline, whereas the final joint refinement stage consistently elevates BCR-only, GSR-only, and full BCGFMamba variants to a similar performance range around 0.8275–0.8278. Together with the updated qualitative comparisons, these results support the practical value of lightweight modular refinement around the bottleneck and decoder pathway for postoperative residual tumor segmentation. Future work will extend the evaluation with broader cross-dataset validation, boundary-sensitive metrics, and efficiency analysis.

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